Lesson 2.2 - Relative Frequency Distribution Table

> **Video:** How To Make a Relative Frequency Distribution Table by The Organic Chemistry Tutor **Chapter:** 2 - Graphs and Tables

Idea

Relative frequency is the **fraction** (or percentage) of the data taken up by each value:

$$\text{relative frequency} = \frac{f}{\sum f} = \frac{f}{n}$$

Steps

1. Make a frequency table: **V** (value) and **f** (frequency).
2. Add the frequency column to get n.
3. Add a third column **RF** = f/n.
4. **Check:** the RF column must add up to **1** (or 100%).

Example

Data (20 values): 2, 5, 3, 4, 7, 2, 5, 8, 3, 2, 2, 3, 2, 5, 5, 4, 8, 5, 2, 8

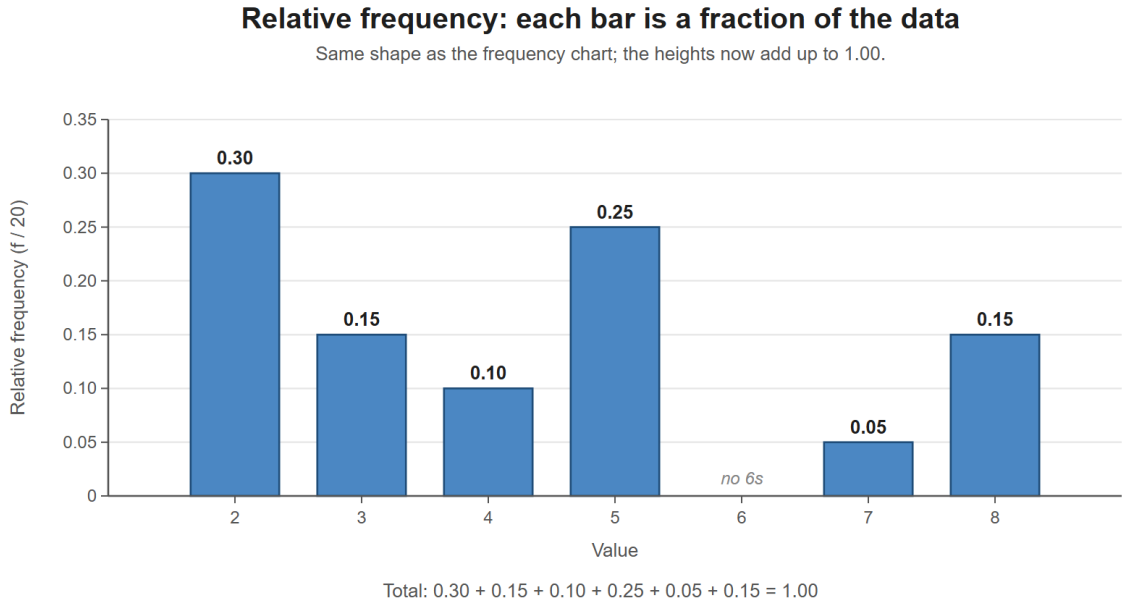
V	f	RF = f / 20
2	6	0.30
3	3	0.15
4	2	0.10
5	5	0.25
7	1	0.05
8	3	0.15
Total	**20**	**1.00**

There is no 6 in the data, so 6 is left out of the table.

Reading the table:

- The relative frequency of the value **4** is **0.10**, so 10% of the data are 4s.
- The relative frequency of the value **8** is **0.15**, so 15% of the data are 8s.

Check: $0.30 + 0.15 + 0.10 + 0.25 + 0.05 + 0.15 = 1.00$ ✓



The relative frequency chart has the same shape as a frequency chart; only the vertical scale changes. Every bar is a fraction of the 20 values, so the heights add up to exactly 1.00.

Key takeaways

- RF = frequency ÷ total frequency.
- RF values are between 0 and 1 and always **sum to 1**. If they don't, recount.
- Multiply by 100 to read RF as a percentage.
- The next step is the **cumulative relative frequency** (Lesson 2.3).